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Phishing Email Detection using Persuasion Cues

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Abstract—Phishing is an attempt to acquire sensitive information from an unsuspecting victim by malicious means. Recent studies have shown that phishers often use persuasion techniques to get positive responses from the recipients. Still missing from this literature are studies assessing effectiveness of persuasion cues in phishing email detection. Specifically focusing on gain and loss persuasion cues, we address the following research questions: In detecting phishing emails, (1) how effective are the gain persuasion cues, (2) how effective are the loss persuasion cues, and (3) how effective is an integrated model of gain and loss persuasion? In order to address the research questions, we create three machine learning models, with relevant gain persuasion cues, loss persuasion cues, and combined gain and loss persuasion cues respectively, and compare the estimates with a baseline model that does not account for the persuasion cues. The results show that the three phishing detection models with relevant persuasion cues significantly outperform the baseline model by approximately 5% to 20% percent in terms of F-score, thus representing reliable methods for phishing email detection. The objective of this study is to develop anti-phishing methods that utilize persuasion cues for detecting phishing emails. Such research is useful because a deeper understanding of persuasion cues can inform the design of effective countermeasures for detecting and blocking phishing emails.

Index Terms—Security and protection, Social science methods or tools, Social technologies

1 Introduction

PHISHING is an act of social engineering to obtain information from an unsuspecting victim [1], [2]. It involves an attacker who generally masquerades as a legitimate institution [3] to trick users into disclosing sensitive information that can later be used in fraudulent activities [4]. More than 100,000 Internet users around the world are subjected to phishing attacks daily[1]. The Gartner Group has estimated billions of dollars in losses from phishing attacks each year [5]. So it comes as no surprise that the literature focusing on phishing detection has gained significant attention in recent times [6].

Prior literature in phishing detection has focused on structural characteristics of phishing messages. For instance, they have focused on classifying phishing emails based on manipulation of senders' IP addresses, URLs, or domain names [7]. Researchers have also examined the content of phishing messages focusing on cues such as presence of logos and the presence of spelling and grammatical mistakes [8], [9]. Recent literature in phishing detection has focused on (deep) mining of content [10], [11], while relying less on theories to explain the detection process for interpretability reasons.

As phishers fabricate phishing content in different ways, the data-driven approaches have less chances of being generalizable. This problem is further exaggerated when phishers influence victims' responses while maintaining authentic and legitimate appearances, which makes it difficult for filters to classify phishing content as fraudulent [12], [13], and users remain vulnerable to phishing messages. Thus, flagging emails as phishing content is one task that many security analysts in security operation centers have to do manually. Therefore, automated, explainable approaches that can provide valuable insights for phishing detection as well as for qualitative and quantitative analysis of phishing data are required.

Towards that end, this paper provides actionable intelligence about emerging phishing attacks (i.e. cyber threat intelligence) by infusing the explainability informed by theoretical perspectives from the social science and psychology literature for effective phishing detection. In particular, it examines the effectiveness of persuasion cues for phishing email detection or design of phishing email filters. Persuasion is a technique of influencing people to grant someone's request [14]. Persuasion cues are signals within phishing emails through which phishers attempt to persuade and influence readers. One example is an email requiring the recipient to reap the reward of winning a lottery by clicking on a link within the email.

Recently, researchers have investigated the role of persuasion cues in phishing susceptibility and victim behaviors [15]. We extend this investigation by gauging the effectiveness of persuasion cues for phishing email detection. Specifically, we ask the research question: how effective are persuasion cues for phishing email detection? In order to address this question, we take a text mining approach to explore two types of persuasion cues, namely gain and loss persuasion cues, within the phishing emails. Utilizing phishing data collected from an anti-phishing service Millersmile[2], we create a machine learning solution that takes into account these persuasion cues within phishing emails for phishing detection and examine its effectiveness.

The contributions of this study are three-fold: (1) to

[1] Kaspersky (2013). The Evolution of Phishing Attacks. Retrieved from http://media.kaspersky.com/pdf/kaspersky_lab_ksn_report_the_evolution_of_phishing_attacks_2011-2013.pdf

[2] http://www.millersmiles.co.uk/

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characterize gain and loss persuasion cues by creating computer-measurable features, and (2) to assess the effectiveness of persuasion cues in phishing email detection, and (3) to develop an IT artifact for phishing email detection using gain and loss persuasion cues. Such research is important because a deeper understanding of phishers' tactics can inform the design of effective countermeasures that directly address security concerns and can help in detecting and blocking phishing messages [16]. This research can help in understanding the phishing detection system using computational and human factors. The utilization of the theoretical perspective can help produce explainable models that can provide interpretations of black-box models (NISTIR 8312[3]).

The rest of this paper is organized as follows: In the next section, we introduce the literature on phishing and persuasion. We then discuss our methodology and the phishing detection model. Subsequently, the evaluation process is introduced, and results are discussed. In closing, we detail limitations and future possibilities.

2 Background

We first discuss approaches for phishing countermeasures and detection, and then focus specifically on phishing persuasion.

2.1 Phishing Countermeasures

To launch a phishing attack, phishers must create a fake website and send emails to attract visitors [17]. Prior literature on phishing detection has focused on content-based approaches such as web page text, code, images, URLs, and link information [12]. [18] provide a comprehensive review to identify three general categories of phishing countermeasures: rule-based, classifiers and manual effort. Mechanisms in the rule-based category utilize keywords, syntax checks, sender address, URLs, etc. to detect phishing emails [19]. In the classifier category, statistics, pattern matching and machine learning is used to create classifiers [20]. In the manual effort category, end users identify phishing emails. This is considered undesirable given that humans are the weak link in the security chain [21]. A detailed review can be found in [22], [23].

Table 1 shows prior literature on phishing detection focusing on content-based approaches [12]. Prior work has utilized natural language processing (NLP) to detect benign or phishing emails [21], [24], [25]. [26] have employed a t-test and WordNet to select features from phishing emails for detection in an attempt to provide structure to the feature selection problem, which previously had been ad hoc [21]. Other works, such as [27], have presented solutions to decide whether communication is deceptive. The application of NLP techniques to phishing detection has hinged on linguistic features, such as commonly known phishing words, proper noun organization names and their types, structural features in the email, etc. [21], [26].

However, recent literature has pointed to drawbacks of NLP in phishing email detection in that the use of synonyms and sentence construction are difficult to find by NLP [21]. It has also identified problems with the classifier category in general, because machine learning methods mainly rely on feature engineering to generate features represented in the emails, which may require heavy manual labor and domain expertise [10]. To solve these problems, this literature has resorted to deep learning techniques. [11] have proposed word embedding and convolutional neural network (CNN) to build a phishing email detection model. [10] have proposed convolutional neural networks (RCNN) model with multilevel vectors for phishing email detection model.

TABLE 1
Content-based Features in Phishing Detection

Paper	Phishing detection cues	Algorithm
Chandrasekaran et al. (2006)	Fake domain, fake pages, URL, fake (spoofed) names, using IP address rather than URL, keywords	Support Vector Machine
Wang et al. (2009)	Email layout, sender's address, bad grammar, embedded URLs	NA
Form et al. (2015)	Blacklist words in subject and content, IP address, features of URL (IP address, symbols, dots in URL), sender, domain, fraudulent hyperlink	Support Vector Machine
Marchal et al. (2014)	URL, IP address, domain	Random Tree, Random Forest, Decision Trees, Support Vector Machine
Kathirvalavakumar et al. (2015)	Domain, URL, size of email	Neural Network
Zielinska et al. (2014)	URL, bad grammar, warning messages	NA
Smadi et al. (2015)	Different domains, IP address, symbols & dots in URL, Hyperlink, email headers, message content, attachments, JavaScript popups	Random Forest, Decision Trees, Multilayer Perceptron, Support Vector Machine, Logistic Regression, Naïve Bayes
Tayal & Ravi (2016)	Features of URL – IP Addresses, dots, slashes, other symbols SSL Certificate, etc.	Associative Classification
Wanawe et al. (2014)	URL, HTML structure of the webpage, IP address	K-Means, Naïve Bayes
Gowtham & Krishnamurthi (2014)	Login form identity, age of form, IP address, domain name (credibility, age of domain)	Support Vector Machine

There are two important gaps in the phishing detection literature. First, a vast majority of the literature in phishing detection has focused on data-driven approaches including mining of phishing content [23]. There is limited research relying on theoretical explanations of the detection process for interpretability reasons. Second, despite the extensive attention being paid to technical solutions to phishing, "organizations remain vulnerable to users receiving and responding to phishing messages" [15]. Phishers broadcast manipulative content through emails, text messages or instant messages maintaining authentic and legitimate appearances, which makes it difficult for users as well as filters to classify phishing content as fraudulent [12], [13]. Phishers fabricate phishing emails using persuasion tendencies that can evade even the sophisticated email filters [7], [28]–[30].

[3] Four Principles of Explainable Artificial Intelligence by National Institute of Standards and Technology (NIST). Retrieved from https://www.nist.gov/system/files/documents/2020/08/17/NIST%20Explainable%20AI%20Draft%20NISTIR8312%20%281%29.pdf

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2.2 Phishing Persuasion

Phishing messages are usually sent to the receiver with intent to mislead them [31]. Phishers tailor their messages to look like a legitimate email [32]. User decisions to click on a link or open an attachment in an email might be influenced by how well a phisher can persuade the user.

Using persuasion techniques, phishers can motivate individuals to comply with the instructions in a message by providing an opportunity or value. For example, phishers may word the message with a suggestion that the recipient has won a lottery, or has been selected to participate in a money-making venture, or has been offered a fortune [33]. According to [14], there are six basic tactics of persuasion: reciprocity, consistency, social proof, likeability, authority and scarcity.

Reciprocity is the tendency to repay in kind, return the favor or adjustment to a request [34]. Consistency is a public commitment where people commit to the decision they have made [35]. Likeability is an act of establishing trust and credibility. Social proof is the behavior of people following their peer group, role models or important others. While it is human nature not to question authority, it can be used to cause fear of punishment, humiliation or condemnation [4]. Scarcity exists when opportunities seem more valuable when they are less available [15]. The emails in our dataset did not provide much information on social proof, authority and scarcity. So we drop these from the analysis in the paper.

Various researchers have examined Cialdini's persuasion tactics within phishing emails, and have identified popular ones (that achieve high user response) within the phishing context. Some of these studies are summarized below (see Table 2). Many of the persuasion tactics, such as scarcity and consistency, have been found popular in the phishing emails [36].

On the one hand, some of Cialdini's persuasion tactics utilize gain framing in order to lure people to comply with a recommended course of action – offering some reward (reciprocity), providing promises (consistency), establishing trust (likeability). These persuasion tactics are designed to excite people by offering a reward i.e. these tactics frame 'gain' from clicking in a phishing email [15], [37]. An example, regarding gain from a reward is shown in Figure 1. Additional examples of gain persuasion cues used by the phishers include: "you have inherited some money", "we have a rewarding deal for you", etc.

On the other hand, some persuasion tactics utilize loss framing in order to scare people with outcomes if they do not comply with a recommended course of action [38]. Such type of persuasion that identifies losses have been shown to arouse recipients' emotional response [39]. Phishers intimidate recipients through threats of losses [40]. An example, regarding loss of service is shown in Figure 2. Loss persuasion cues are intended to scare people by framing a 'loss'. Additional examples of loss-framed cues used by the phishers include: "your service could be suspended!", "your online account is locked", and "your account has been terminated".

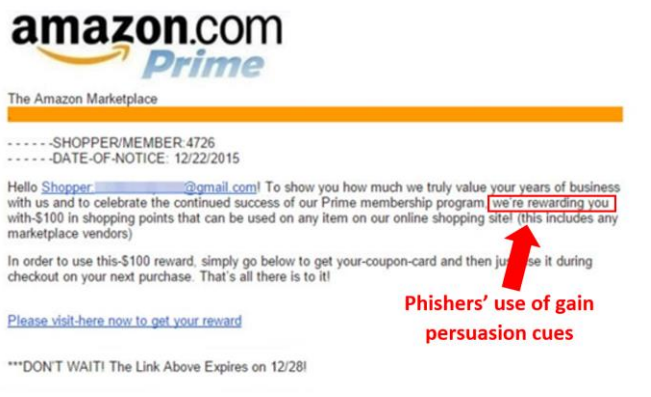


Fig. 1. Strategic Use of Gain Persuasion Cues by Phishers in Emails[4]

There has been anecdotal evidence regarding the increasing use of loss persuasion cues by the phishers within the phishing emails (see [38], [39], [41]). Loss persuasion cues focus on describing the dreadful things that may happen to the recipient. For phishing messages that denote the fictitious loss, users may take actions to control their loss [42]. Performing the actions specified within the email may help the user alleviate outcomes related to the loss, however resulting in the user being phished. Some actions may also be influenced based on the severity of the loss. [43]



Fig. 2. Strategic Use of Loss Persuasion Cues by Phishers in Emails[5]

identify severity in terms of elaborate details, vivid descriptions as well as graphics. In addition, immediacy of the loss may also invoke actions [38].

In line with the above, the research associated with persuasion cues underscores that the expressions of gains and losses exert a strong influence over people's judgments and actions in the context of phishing. The major premise is that successful phishing attacks take advantage of the human tendency to make judgments based on message framing that suggests that the recipient gains or loses something of value.

While many studies have analyzed persuasion cues in phishing emails and some studies have shown the possibility of automatically detecting persuasion, the effectiveness of persuasion cues for phishing detection has been subject of limited investigation. In order to address this, we create an integrated framework for developing computer-measurable features of gain (through reciprocation, consistency and likeability) and loss (through loss, severity

[4] Adapted from https://www.virtualarmour.com/dont-let-phishing-scams-catch-you-unaware/

[5] Adapted from http://blogen.stickypassword.com/3-phishy-things-to-watch-out-for/

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and immediacy) persuasion for the purposes of phishing email detection.

3 Methodology

In this section, first we discuss the data collection methodology in detail. Then we describe our datasets and the processing required to transform free-form email data into a form suitable for rigorous evaluation Next, we specify our coding scheme and coding reliability. Finally, we present details of the phishing detection model.

3.1 Data Collection

We collected phishing data from an anti-phishing service Millersmile[6] and legitimate emails were downloaded and extracted from the Enron corpus.

Millersmile maintains an archive of more than 2 million phishing emails. We collected data about phishing emails, recording the date reported, risk level, email subject, apparent sender, return address, email format, URL of web content, anchor text of URLs, location, scam number, comments, contents and website. For this research, we collected 17,902 unique phishing email scams from November 2007 to April 2015. We also collected an additional 521 unique phishing emails for pilot coding purposes.

The Enron corpus is a large database of over 600,000 emails generated by 158 employees of the Enron Corporation, which was acquired by the Federal Energy Regulatory Commission during its investigation. This corpus is one of the publicly available mass collections of legitimate emails. We collected 19,153 unique emails as the legitimate emails. We also collected an additional 508 unique legitimate emails for the purposes of pilot coding.

3.2 Data Preprocessing

In this subsection, we discuss the preprocessing steps that were performed on the dataset. Preprocessing is important given the noisy nature of email content. We removed duplicates from each dataset. We preprocessed the email content through three steps: text cleaning, text preparation and text refinement [44]:

Text cleaning consists of removing duplicate emails that are sent/forwarded to the large number of recipients typically in an organizational setting. Text preparation includes segmenting the email text to tokens using NLTK tokenizer package[7]. Grammar is usually ignored in the collection and construction of tokens. Text preparation also consists of removing non-alphanumeric characters (e.g. '@' or '#') and stop words (such as 'the'). Text refinement involves conversion to lower case and stemming. Stemming is the process of producing morphological variants of a root/base word. For example, stemming reduces the words "retrieval", "retrieved", "retrieves" to the stem "retrieve". The pre-processed data is then supplied to the phishing email detection model.

TABLE 2
Cialdini's Persuasion Principles in Phishing Context

Study and Country	Motivation	Popular Principles
Akbar (2014); Netherlands	Understanding the characteristics of phishing emails Present an analysis of persuasion techniques in phishing emails	Authority Scarcity, consistency and likeability (when consider target types and reason typ
Ferreira et al. (2015); Luxemborg and England	Identify elements which reflect the effectiveness of phishing, and manually quantify them within a phishing email sample	Likeability
Valecha et al. (2015); United States	Examine language of persuasion within phishing emails	Reciprocity and scarcity
Butavicius et al. (2016); Australia	Understanding users' judgments of how safe it is to click on a link in genuine, phishing or spear-phishing emails	Authority
Zeilinska et al. (2016); United States	Examine the persuasion principles present in emails available in the United States over a period of five years	Consistency and scarcity

3.3 Persuasion Cues

This paper investigates the effectiveness of gain and loss persuasion cues for phishing email detection. This requires us to create computer-measurable features of gain and loss persuasion cues. We accomplished this in three steps: 1) human coding N=1029 emails for generating gain and loss persuasion labels, 2) training a machine learning model on human coded emails for predicting persuasion labels on the larger dataset of N=37,055 emails, and 3) extracting lexical features from email content for classifying N=37,055 emails as phishing or legitimate emails. These steps are detailed as follows:

1) For coding persuasion cues, we utilized the human coding approach. In order to code gain and loss persuasion cues, we followed the steps for content coding and analysis suggested by [45]. Since gain and loss persuasion cues are coded based on certain tactics that can result in gain (for example, click on the link for $100 gift card) or loss (for example, limited time offer will expire soon), human coding is suitable to derive objective measures for gain and loss persuasion cues. Through human coding, we obtained persuasion labels of gain (reciprocation, consistency and likeability) and loss (loss, severity and immediacy) cues for the pilot dataset of N=1029 emails (521 phish and 508 legitimate). For details on human coding of gain and loss persuasion cues, refer to Appendix A.

2) In order to obtain persuasion labels for the full dataset of N=37,055 emails, we resorted to machine learning wherein the human coded persuasion cues from the pilot dataset were fitted into a bidirectional long short-term memory model (Bi-LSTM) to generate prediction of persuasion labels of gain and loss for the full dataset. In simple words, we trained a Bi-LSTM model on the pilot dataset to obtain gain and loss persuasion predictions for the full dataset. For details on the Bi-LSTM model for predicting gain and loss persuasions, refer to Appendix B.

3) For extracting lexical features of persuasion, we selected Word2Vector (W2V) model. W2V utilizes continuous or skip-gram algorithm, which predictively learns the

[6] http://millersmiles.co.uk/

[7] http://www.nltk.org/api/nltk.tokenize

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word embedding or the numeric vector representation of words. This trained neural network model understands the context of each word in the dataset. As a result, W2V has a superior performance compared to deterministic models such as Bag of Words (BOW).

In this paper, we have created 4 W2V models: 1) W2V with relevant gain persuasion cues (W2V_GAIN), 2) W2V with relevant loss persuasion cues (W2V_LOSS), 3) W2V with relevant gain and loss persuasion cues (W2V_GAINLOSS), and 4) W2V without any relevant persuasion cues (W2V_NONE). W2V_GAIN and W2V_LOSS were implemented on phishing and legitimate emails with any of the (predicted)[8] gain persuasion cues and any of the (predicted) loss persuasion cues respectively. W2V_GAINLOSS was implemented on emails with any of the (predicted) persuasion cues. W2V_NONE was implemented as a baseline model on emails because it did not contain any (predicted) persuasion cues. We ensured that the vocabulary for the W2V model without persuasion cues did not overlap with that of the W2V models utilizing the persuasion cues. The classifiers utilized with this method appear below. Table 3 shows some of the top features in each of the W2V models.

3.4 Descriptive Summary

There were 37,055 email messages (17,902 phish and 19,153 legitimate) in the dataset. Out of the 37,055 emails, there were 12,911 emails with gain persuasion cues (11,194 emails with reciprocation persuasion, 190 emails with consistency persuasion, 2049 with likeability persuasion) and 8,092 emails with loss persuasion cues (7,901 emails with loss persuasion, 2,259 with severity persuasion, 798 emails with immediacy persuasion).

3.5 Validation Method

The effectiveness of persuasion cues for phishing email detection was evaluated by a random sub-sampling validation method. This method uses random partitions of training data and test data. We resorted to the standard practice of k-fold cross validation [46], wherein a small portion of the dataset is used without the class labels for testing purposes, and this is repeated for 'k' separate portions of the dataset. While there are many ways of choosing this subset [47], we utilize a method whereby each subset consisted of consecutive rows from the main dataset. For the validation process, we used k=10 to create an 90/10 split (this is subsequently elaborated in section 3.8).

TABLE 3
SAMPLE OF TOP FEATURES IN WORD2VECTOR (W2V) MODELS

W2V	Sample Features (among others)
W2V_GAIN	Free, Money, Secure, Restore, Fund, Service,
W2V_LOSS	Failure, Irregular, Locked, Incorrect, Invalid, fraud, Blocked, restricted, Limited
W2V_GAINLOSS	Password, Holder, Payment, Attempts, Banking, Fraudulent, Copyright, Prevent, Access
W2V_NONE	Cost, Tax, Technology, Letter, Limit, Execute, Exchange, Draft, Agreement, Legal, Action

3.6 Performance Metrics

The confusion matrix is widely used in the literature to evaluate prediction algorithms [48]. Confusion matrix is a relevant measure to use to evaluate the performance of the phishing detection model. Within the confusion matrix, the lower the number of misclassifications, the better is the performance. The extent of misclassification is assessed by the number of Type I and Type II errors. We developed a confusion matrix to compare the results of the phishing detection model. The confusion matrix allows us to determine the percentage of correctly classified and misclassified phishing emails.

The evaluation metrics employed in the experiments included those used in prior research: overall accuracy, precision, recall and F-score [12]. With the help of the confusion matrix, we define the evaluation metrics as follows: We define accuracy as the proportion of the total number of emails that were correctly classified. Precision is the proportion of the predicted phish that were correctly classified. Recall is the proportion of actual phish that were correctly classified. F-measure is the geometric mean of precision and recall [49].

3.7 Machine Learning Models

Since we utilize the structural properties (pointing to persuasion tactics) of emails, we tested and evaluated the performance and applicability of baseline models that use explainable textual features. More specifically, we used supervised machine learning to show the efficacy of persuasion techniques in classifying phishing emails.

According to [50], there are three traditional approaches to constructing a classification[9] model: (1) create different subsamples from a training set, build a classifier on each set, and compare the estimates, (2) create different subsets of features to build classifiers, and compare their estimates, and (3) manipulate the output targets to build multiple classifiers and compare their estimates [51]. In this study, we use the second approach to build a detection model as suggested by [51]. We choose the second approach because we wanted to test the effectiveness of persuasive features. Using the second approach, we can manipulate the set of input features, namely persuasive vs. non-persuasive, available to the learning algorithms and compare their estimates. If the estimates of the learning algorithm utilizing persuasive features outperform that of the algorithm utilizing non-persuasive features, then we can conclude that persuasive features were effective in phishing email detection. To elaborate, we created four Word2Vector (W2V) models that utilize different subset of features based on gain and loss persuasion cues.

For the purposes of generalizability of our findings, we

[8] Predicted cues are derived from Step 2.

[9] In this paper, we use the term detection and classification interchangeably to decrease monotony.

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utilized four models (commonly used in the literature depicted in Table 1) within scikit-learn, an open source python-based collection of machine learning tools, for phishing email classification: Naïve Bayes, Logistic Regression, Random Forest and Support Vector Machines [52], [53].

(1) Naïve Bayes (NB): NB is particularly suited when the dimensionality of the inputs is high or when input data is large. NB can be useful for exploring data by providing the probability of each input attribute [54].

(2) Logistic Regression (LR): LR is used to estimate the probability of a binary response based on one or more features. It allows one to say that the presence of a feature increases the probability of a given label by a specific percentage [52].

(3) Random Forest (RF): RF is one of the most popular and most powerful machine learning algorithms suitable for large data sets. It constructs a set of decision trees in the training phase, each tree operating on randomly chosen attributes for prediction. During the training phase, several decision trees are constructed (we used 5 trees) which are then used for the class prediction; this is accomplished by considering the voted classes of all the individual trees and the class with the majority vote is the prediction [55].

(4) Support Vector Machine (SVM): SVM is a supervised machine learning model that is mostly used for classification tasks, wherein each data item is plotted as a point in n-dimensional feature space so that it can determine the best hyperplane that divides the classes. SVM is known for its accuracy as well as a linear model for supervised machine learning of textual content [56].

Several researchers have created deep learning models for phishing detection [10], [11], such as Convoluted Neural Networks (CNN; [57]) or Long Short Term Memory (LSTM; [58]). These deep learning models create latent features i.e. automatically generated features that are obtained by matrix (tensor) factorization. Though these latent features can perform well for phishing email detection, they are often difficult to comprehend. In addition, deep learning models present a black-box model for phishing email detection. Owing to the advances in explainability of deep learning solutions for phishing email detection in the recent literature, they can help answer why a particular email is detected as phish. However, deep learning models cannot attribute phishing emails to persuasion tactics within the emails.

3.8 Hyperparameter Tuning

The key part of any machine learning model is to tune parameters for good performance. In order to optimize the hyperparameters for the models, we test the performance of multiple parameters and their combinations to determine the most optimal set of parameters. There are several methods for searching hyperparameter values, such as grid search, random search, manual search, etc. Grid search is a brute force method that tries all the possible combination of hyperparameter values. Random search is an iterative method of search for possible combinations randomly. Manual search, is a manual process trying different combinations based on model application and understanding [59]. We resorted to the random search technique for hyperparameter tuning.

This was performed using 10-fold cross validation in the following steps: 1) We split the dataset 90/10 for training set (90%) and hold out set as a test (10%). 2) Subsequently, we utilize random search on the training set by splitting it 90/10 for training set (90%) and validation set (10%). 3) Then, we search for the optimal hyperparameters. 4) Next, we obtained the best performing model by fitting on the whole training set and validating on the validation set. 5) Finally, we tested it on the hold out data in the test set over 10-fold and averaged it to get final estimate of the performance of the model.

We estimated the best model over a range of values for various parameters of the classifiers Naïve Bayes (NB), Logistic Regression (LR), Random Forest (RF) and Support Vector Machine (SVM).

NB classifier was trained with variance smoothing of 1e-07, 1e-08, 1e-09 and 1e-10. LR classifier was trained with maximum iterations of 10, 50, 100, 200 and 1000, with saga, newton and lbfgs solvers. RF classifier was trained with 10, 50, 100, 200 and 1000 estimators, and with minimum impurities of 0, 0.01 and 0.1. SVM classifier was trained with maximum iterations of 10, 50, 100, 200 and 1000 with linear, sigmoid and rbf kernels.

For the four models (W2V_GAIN, W2V_LOSS, W2V_GAINLOSS and W2V_NONE), NB performed best with variance smoothing of 1e-07. For W2V_GAIN and W2V_LOSS models, LR performed best using newton solvers. For W2V_GAINLOSS and W2V_NONE models, LR performed best using the maximum iteration of 200 and 1000 respectively. For W2V_GAIN and W2V_LOSS model, RF performed best using 200 estimators with 0 impurities. For W2V_GAINLOSS models, RF performed best using 0 impurities. For W2V_NONE model, RF performed best using 50 estimators with 0 impurities. For W2V_GAIN, W2V_LOSS, W2V_GAINLOSS and W2V_None, SVM performed best using the maximum iterations of 2000. We also utilized various sizes of vocabulary for the W2V models ranging from 200 to 3000 most frequent words, with increments of 200. The model performs the best using 200 most frequent words in the emails.

TABLE 4

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COMPARISON OF CANDIDATE MODELS FOR GAIN AND LOSS PERSUASION CUES

	Accuracy	Precision	Recall	F-Score	F-Score Difference (with Baselines)	T-statistic
W2V_NONE: Word2Vector models without any relevant persuasion cues (Baseline)						
Naïve Bayes	78.11%	84.35%	67.13%	74.76%	-	-
Logistic Regression	88.13%	83.61%	93.82%	88.42%	-	-
Random Forest	89.57%	84.84%	95.45%	89.84%	-	-
Support Vector Machine	78.85%	72.19%	91.86%	80.76%	-	-
W2V_GAIN: Word2Vector models with relevant gain persuasion cues						
Naïve Bayes	94.82%	98.69%	90.48%	94.40%	19.64%	t=4020.7; p<0.001
Logistic Regression	96.39%	98.39%	94.07%	96.18%	7.76%	t=2510.6; p<0.001
Random Forest	96.42%	98.53%	93.96%	96.19%	6.35%	t=2100.6; p<0.001
Support Vector Machine	96.52%	98.53%	94.20%	96.31%	15.55%	t=1087.5; p<0.001
W2V_LOSS: Word2Vector models with relevant loss persuasion cues						
Naïve Bayes	91.97%	95.99%	87.02%	91.28%	16.52%	t=2710.7; p<0.001
Logistic Regression	95.89%	98.66%	92.76%	95.62%	7.20%	t=2172.8; p<0.001
Random Forest	95.84%	98.48%	92.83%	95.57%	5.73%	t=1704.4; p<0.001
Support Vector Machine	96.16%	99.02%	92.99%	95.91%	15.15%	t=1059.7; p<0.001
W2V_GAINLOSS: Word2Vector models with relevant gain and loss persuasion cues						
Naïve Bayes	91.91%	95.46%	87.42%	91.26%	16.50%	t=2859.3; p<0.001
Logistic Regression	95.76%	98.78%	92.38%	95.47%	7.05%	t=2095.6; p<0.001
Random Forest	95.53%	98.39%	92.25%	95.21%	5.37%	t=1700.5; p<0.001
Support Vector Machine	95.97%	99.09%	92.52%	95.69%	14.93%	t=1050.2; p<0.001

4 EVALUATION OF PHISHING DETECTION MODEL

In this paper, we posit that there could be persuasion cues within email messages, which may provide valuable hints in determining the legitimacy of the emails. So we evaluate the persuasion cues across the phishing emails by constructing a rich feature set composed of gain persuasion cues of reciprocity, consistency and likeability, and loss persuasion cues focusing on loss, severity and immediacy. We used these features to create a phishing detection model. In evaluating the phishing detection model, first, we discuss the research questions, and then, summarize the results of the evaluation process.

4.1 Evaluation Questions

Three experiments were conducted to evaluate the effectiveness of persuasion cues for phishing email detection with respect to the research gaps identified earlier. In these experiments, we tested if phishing detection models with gain and loss persuasion cues enable us to detect phishing emails in a more general fashion. A brief overview of the experiments and associated research questions is as follows.

In Experiment 1, we compared the effectiveness of the phishing detection model that utilizes gain persuasion cues for phishing email detection, and investigated the following research question:

RQ1: How effective is the phishing detection model with relevant gain persuasion cues – reciprocity, consistency and likeability – compared to a baseline model without any persuasion cues, in terms of F-scores?

Experiment 2 compared the effectiveness of the phishing detection model that utilizes loss persuasion cues to shed light on the following research question:

RQ2: How effective is the phishing detection model with relevant loss persuasion cues – loss, severity and immediacy – compared to a baseline model without any persuasion cues, in terms of F-scores?

Experiment 3 compared the effectiveness of the phishing detection model that integrates gain and loss persuasion cues. The key research question is as follows:

RQ3: How effective is the phishing detection model that integrates relevant gain and loss persuasion cues, versus a baseline model without any persuasion cues, in terms of F-scores?

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Collectively, Experiments 1–3 were intended to demonstrate the overall effectiveness of the persuasion cues in phishing email detection.

4.2 Evaluation Results

In this subsection, we elaborate upon the three aforementioned experiments that were conducted to evaluate the performance of the phishing detection models with gain and loss persuasion cues based on the accuracy, precision, recall and F-score. We summarize the results of the evaluation process (see Table 4) and discuss the effectiveness of gain and loss persuasion cues in phishing email detection. In general, the phishing detection models with persuasion cues outperform the baseline model as explained below:

Experiment 1

As presented in Table 4, the phishing detection models with relevant gain persuasion cues outperformed the baseline models without relevant gain persuasion cues in terms of accuracy, precision, recall and F-score. The performance gains in terms of F-score was approximately 20, 8, 6 and 16 percent over the baseline models. The results were significant at $p < 0.001$. The results suggest that the phishing detection model with relevant gain persuasion cues represents a reliable method for phishing email detection.

Experiment 2

Table 4 shows that phishing detection models with relevant loss persuasion cues provided higher overall accuracy, precision, recall and F-score. The models significantly outperformed the baseline models without relevant loss persuasion cues by approximately 17, 7, 6 and 15 percent in terms of F-score at $p < 0.001$. This implies that loss persuasion cues show a very consistent performance in phishing email classification (F-scores). By illustrating the consistently effective performance of the loss persuasion cues, the results underscore its enhanced generalizability in phishing email detection.

Experiment 3

In this experiment, we evaluated the phishing detection models with relevant gain and loss persuasion cues in comparison with baselines without any gain or loss persuasion cues for phishing email detection. The performance gains for the phishing detection models, in terms of F-score, was approximately 17, 7, 5 and 15 percent over baseline models. The results were significant at $p < 0.001$. Thus, persuasion-based phishing detection model's ability to better detect phishing emails set it apart from the baseline phishing detection models.

5 Explainability

We depict the gain and loss persuasion cues in the phishing emails. The text associated with phishing emails offers some explanation the email recipients. As shown in Table 5, compared to the legitimate emails, phishing emails are persuasive [36]. Phishing emails are written to persuade readers to click on links by offering gains or inciting loss. The predicted scores confirm that the persuasive models can indeed detect phishing emails.

6 Discussion

This study extends existing security research by presenting a theory-based approach to phishing email detection. To the best of our knowledge, this is one of the early papers that develops computer-measurable features for automated phishing detection utilizing persuasion cues. This paper provides implications for explainable AI (artificial intelligence). [60] provide evidence that "many models being used in explainable AI research are not building on current scientific understanding of explanation" (p. 2-3). To address this, we infuse results from social and psychological research focusing on persuasions into explainable AI (through an understanding of the role of persuasion in phishing detection). In addition, [60] point to the need for models "built on an understanding of explanation" (p. 1). By utilizing persuasion cues, we provide an understanding of explanation into the process of phishing email detection. Towards that end, this paper unpacks "black-box models with simpler interpretable models" [61]. In other words, it provides an explanation of the black-box (phishing email detection model) using the persuasion perspective (interpretable model).

TABLE 5
Illustrated Examples of Persuasive Cues in Phishing Emails

Phishing Emails	Persuasion Cues	Predicted Label
DEAR VALUED CUSTOMER, At Alliance & Leicester, we want you to be confident that banking online with us is safe and secure. For us, this means employing the latest technology and multiple layers of security to safeguard your personal details. Our Technical services Department are carrying out a planned software upgrade for the maximum convenience of the users of online-services of the Alliance & Leicester internet banking...	Reciprocation	W2V_GAIN: Phish W2V_NONE: Legit
Dear Valued Customer: We are unable to activate your banking account due to the resent upgrade of our online service, we are sorry for any inconvinences this might cost you. We advise you to reactive your Abbey National online banking account. As part of our security measures Click here to continue your online banking. NOTE:- Your account might be place on restricted status. Restricted accounts continue to receive payments, but they are limited in their ability…	Loss, Severity	W2V_LOSS: Phish W2V_NONE: Legit
DEAR CUSTOMER, Your online account was flagged, for your security we have placed it on limited access. We have also provided an information list and Challenge Questions to your account verification. Please Sign On to verify your account unto Citibank Online. Sincerely, Citibank International Plc…	Loss, Severity	W2V_LOSS: Phish W2V_NONE: Legit
We are writing to let you know that our monitoring process has temporarily suspended your online access to protect your accounts. This has been done due to the number of incorrect login attempts. At Halifax Bank, we take the job of protecting our customers very seriously, so for your protection we are proactively notifying you of this activity. To restore access to your online banking please click on the link below...	Reciprocation, Loss, Severity	W2V_GAINLOSS: Phish W2V_NONE: Legit

Our research extends understanding of phishing detection in multiple ways. First, within phishing email communication, this study has shown the effectiveness of persuasion cues as a factor for detection of phishing emails. We argue that, because phishers attempt to deceive email recipients by persuading them to click on links or download attachments by utilizing persuasion cues, these cues may

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affect the success of a phishing attempt, and in turn prove useful and effective in phishing email detection.

Second, phishers have been using gain persuasion cues to lure people into complying as well as loss persuasion cues to scare users against non-compliance. While many studies have examined gain persuasion cues in phishing emails, fewer studies have examined them in conjunction with loss persuasion cues. Utilizing data from Millersmiles, we identify that a large proportion of phishing emails utilize one or more measures of loss. As phishers utilize various dimensions of loss to invoke action by the users, it is important for researchers to investigate the loss persuasion cues in the context of phishing emails.

Third, while gain persuasion cues have been a focus of numerous studies [15], [62] albeit in investigating phishing susceptibility, in this paper, we examine the effectiveness of an integrated model comprising of gain and loss cues in classifying phishing emails. Based on an analysis of several thousand phishing and legitimate emails, the results indicate that the integrated model of gain and loss persuasion cues performs significantly better than the baseline models used in classification of phishing emails. By integrating cues regarding (a) the loss due to non-compliance with (b) the gain for compliance, we introduce a novel angle to detect phishing email messages.

7 Practical Implications

This paper makes a methodological contribution by proposing an approach for identifying persuasion cues and assessing the effectiveness of these persuasion cues. Organizations could design anti-phishing strategies by understanding the phishers' persuasion cues (both gain and loss cues).

From a practical standpoint, models utilizing persuasion cues could be valuable in identifying what kinds of phishing emails should be prioritized[10]. Those emails with persuasion cues can facilitate additional analysis of emails by security analysts. This may be useful in developing automated and explainable techniques for phishing email detection.

This paper provides clear interpretation of why an algorithm detected an email as a phishing email. In this way, interpretability can improve phishing detection model's performance and develop algorithm trustworthiness. Ultimately, explainable models can encourage security analysts in leveraging Artificial Intelligence (AI) for phishing email detection.

8 Conclusion

This study discusses an anti-phishing method that utilizes persuasion cues for detecting phishing emails. It investigates the effectiveness of persuasion cues in phishing email detection. In particular, it examines gain, loss, and combined gain loss persuasion cues for detecting phishing emails.

This study has the following limitations. We coded some variables in binary data form; there could be information loss during coding and analysis. It was, however, an inevitable choice since coders manually read and coded data from emails. Due to the laborious process, only four master's students coded the persuasion cues.

There are many possible extensions to this work. One future direction for this work would be to incorporate additional loss characteristics such as preventable, mitigatable, observable losses etc. into the model. Another future work would be to investigate whether two-way and even three-way interactions can be modeled, for example, the interaction of gain and loss persuasion techniques. Future work could also comment on the performance of the machine learning models with relevant persuasion cues based on different data preprocessing techniques besides removing duplicates, stop words, etc.[11]

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BIOGRAPHY

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